Strata Cobot

Autonomous ore sorting and haulage with two robots in ROS 2

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Robotics course project
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Outline

1. Introduction
2. The manipulator
3. The mobile robot
4. Results
5. Conclusion

Two robots, one mining mission

1. **eBot crosses the site**
   plans its own route around the rocks
2. **eBot takes on the ore**
   collects an ore package at the drop pose
3. **Arm sees each ore**
   RGB-D camera finds every ore in 3D
4. **Arm sorts it**
   magnet picks each ore and drops it in its container

Three minerals, three containers

- Azurite (blue)
- Malachite (green)
- Vanadinite (orange)

Our constraint

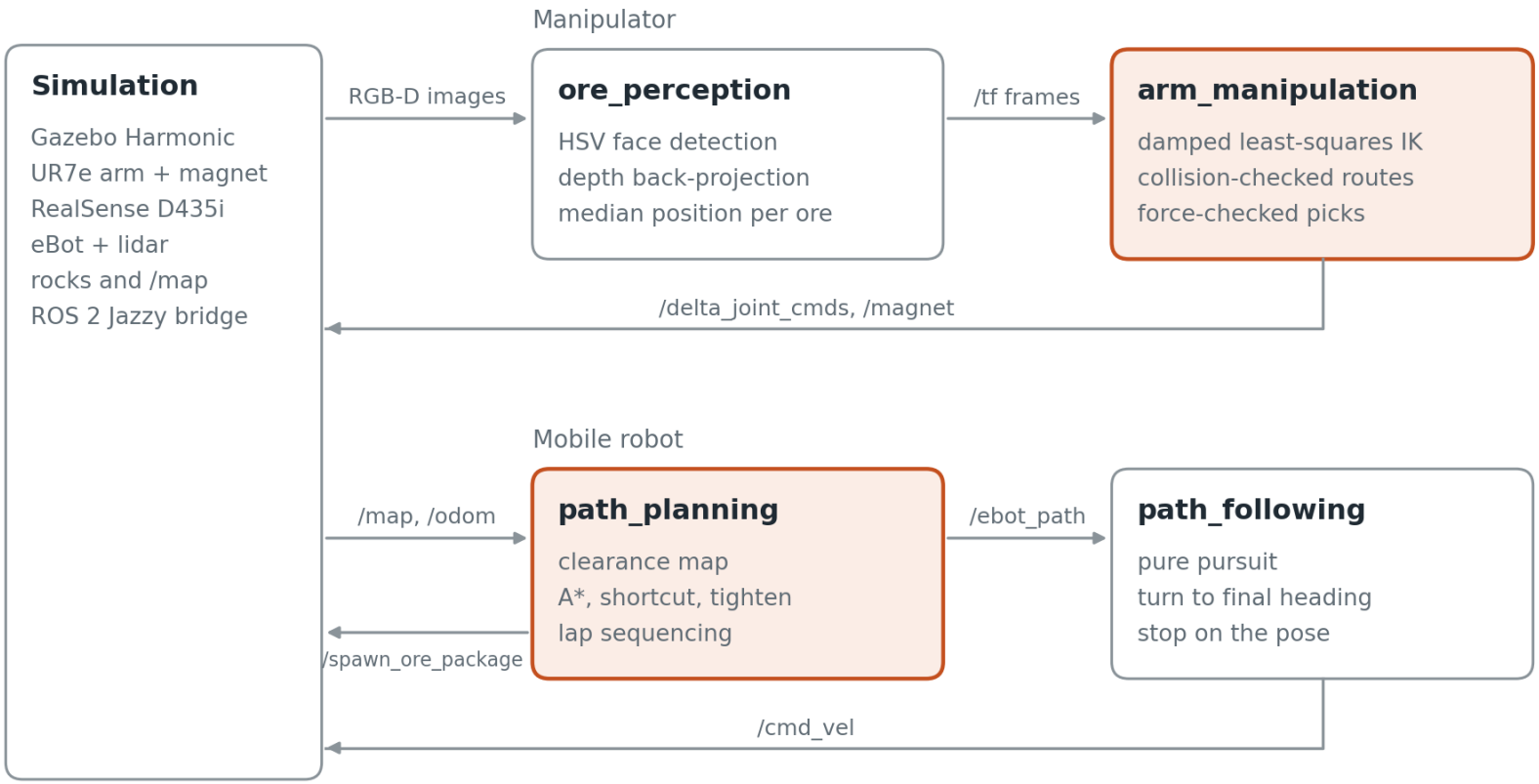
No MoveIt 2, no Nav2: every planner and controller written from scratch.

Four goals, each with a number to check

Subsystem	Robot	Goal	Measured by
Perception	Arm camera	Locate every ore in 3D and name its mineral	error vs. ground truth
Manipulation	UR7e arm	Pick each ore, drop it in its container	ores sorted correctly
Path planning	eBot	Collision-free path for each leg	length vs. shortest
Path following	eBot	Track each path, stop on each pose	tracking error

Four nodes, two sense–plan–act pipelines

Each robot runs a sense, plan, act pipeline of two nodes

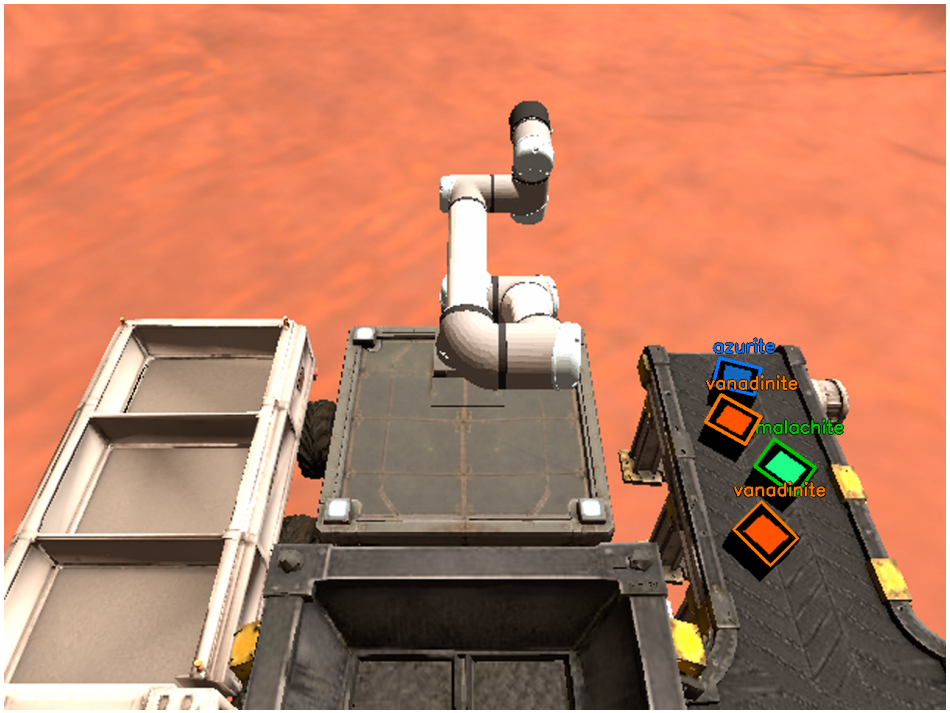


The arm accepts only joint velocities and the eBot only velocity twists, so every node closes its own control loop at 20–50 Hz.

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Find each ore by shape, name it by a colour vote



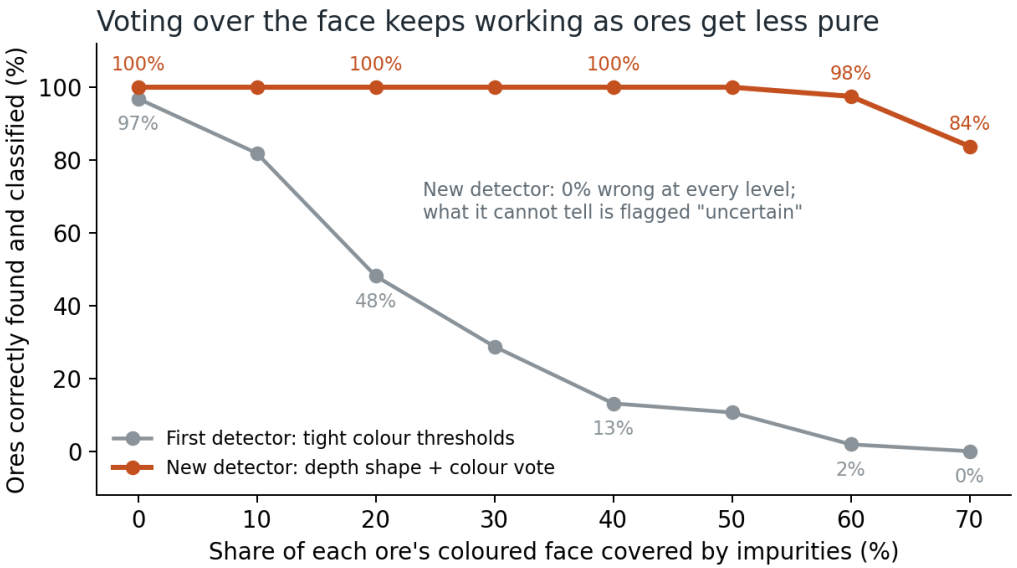
Where: depth

An ore's top face is the only surface 66–86 mm above the belt. Centre of a rectangle fitted to it: **2–3 mm** error.

What: colour vote

Share of the face in each mineral's colour (HSV hue). The winner needs $\geq$3% and 2$\times$ the runner-up, else *uncertain*. Majority vote over 20 frames.

Real ores are rarely one colour, so we vote



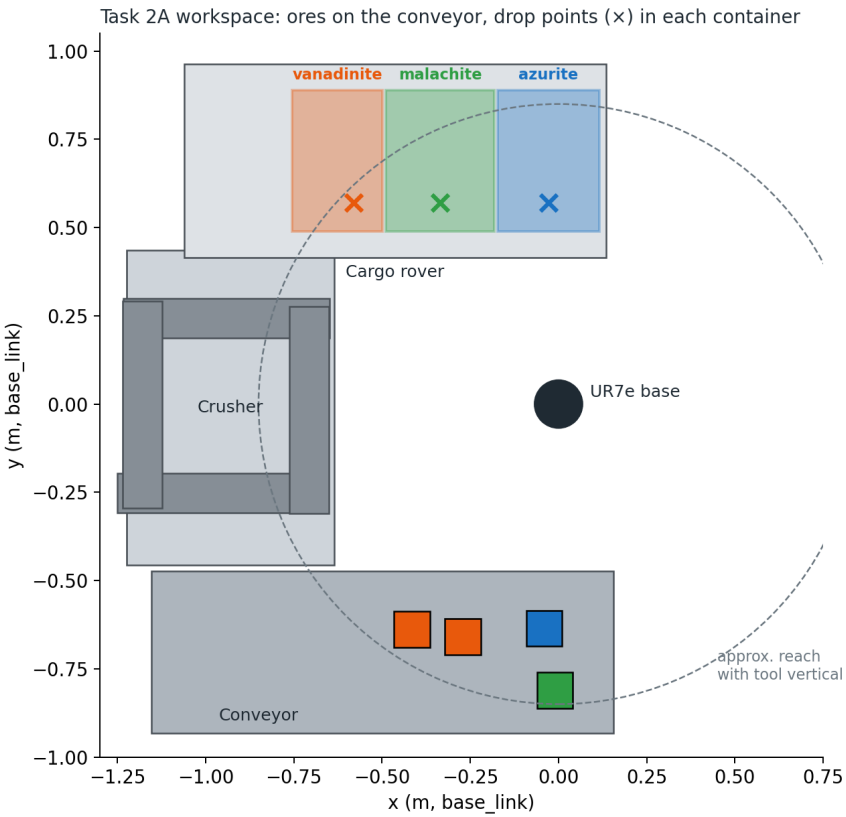
Impurities painted into a real camera frame, varied lighting; 160 ores per level.

180 real museum photos (Wikimedia Commons); calibrated on 90, tested on 90 others:

Held-out	**Tight**	**Vote**
Azurite	10%	37%
Malachite	7%	80%
Vanadinite	17%	93%
All	**11%**	**70%**

Azurite is hardest: very dark, often crusted with green malachite.

The containers were found by measuring the rover mesh



- Not given: parsed the cargo rover's collision mesh (GLB) into `base_link`
- Gazebo's default view: vanadinite nearest, malachite middle, azurite farthest
- Vanadinite centre is **0.93 m** away: out of reach with the tool vertical
- So each ore is released at the nearest reachable point inside its container

Every pick is confirmed by the force sensor

1. **Hover** 10 cm above the face, tool vertical
2. **Align** the last millimetres on `/tcp_pose_raw`; magnet on
3. **Descend** at 4 then 1.2 cm/s; stop at the face or at 2.5 N
4. **Lift and check**: 1.47 N means one 0.15 kg ore is held; under 0.8 N, retry
5. **Drop**: carry with the ore in the collision model, release

81 mm
magnet disc on a 101.6 mm face

1.47 N
one ore held

0
safety stops in all trials

A purpose-built IK solver plans 300× faster

Damped least squares, 5 constraints (position + vertical tool axis)

$$\Delta q = J^T \left(J J^T + \lambda^2 I \right)^{-1} e, \qquad \lambda = 0.02$$

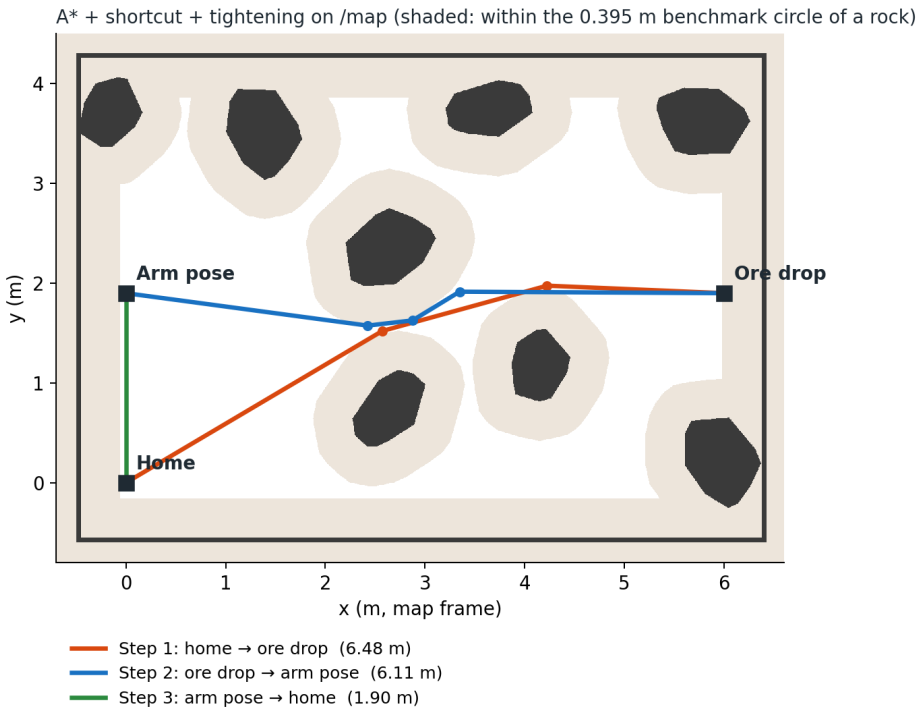
- Rotation about the vertical is free: the magnet face is round
- Safety checked once on the result: joint limits, 3 cm clearance, singularity ($\sigma_{\min} \geq 0.04$), tilt $\leq 1^\circ$
- Every solved pose is reused as a seed, so the arm always finds its way back
- Straight vertical moves by resolved-rate control: $\dot{q} = J^{+}[v;\, \omega]$

SLSQP optimiser: **minutes** per plan $\longrightarrow$ DLS: **0.1–0.4 s**

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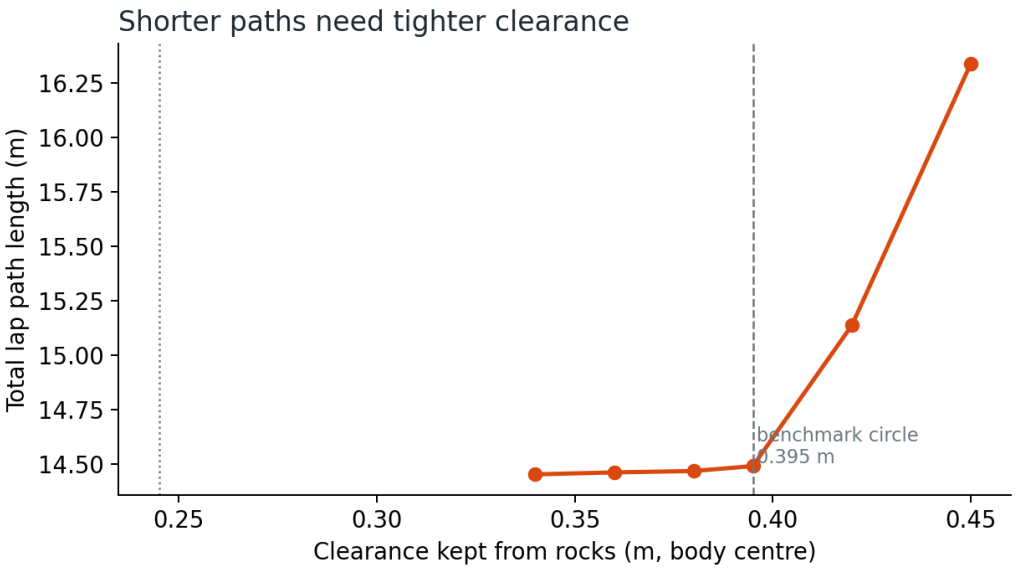
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A*, then shortcut, then tighten



1. **Clearance map**: distance transform of the 5 mm `/map`
2. **A***: 8-connected, 5 cm grid, only cells with enough clearance
3. **Shortcut**: jump to the furthest visible point
4. **Tighten**: pull each corner in until the legs just clear

Below 0.40 m clearance, paths stop getting shorter



- Reference path: the robot as a **0.395 m** circle
- Driving straight, only the **0.245 m** half-width faces a rock
- We take the safest clearance (0.40–0.34 m) no longer than the reference
- At 0.38 m: 13.5 cm each side; tracking error $\leq$ 9 cm

Pure pursuit follows each path

- Steer to a point 0.30 m ahead on the path
- Turn rate $\omega = \mathrm{clip}(2.2\, e_\theta, \pm 1)$; speed $v = 0.35 \cos^2 e_\theta$
- Above 40° of heading error: turn on the spot first
- At the end: turn to the path's final heading and stop
- The lidar only slows the robot; the path is already clear of every rock

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Every ore sorted, every lap completed, no collisions

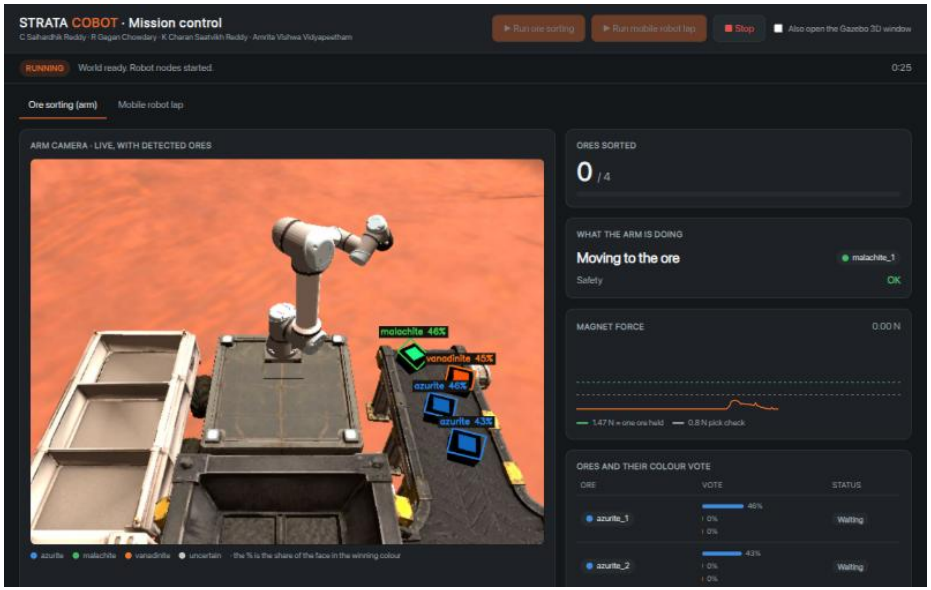
16/16
ores sorted correctly, 4 trials

1.00
best path efficiency

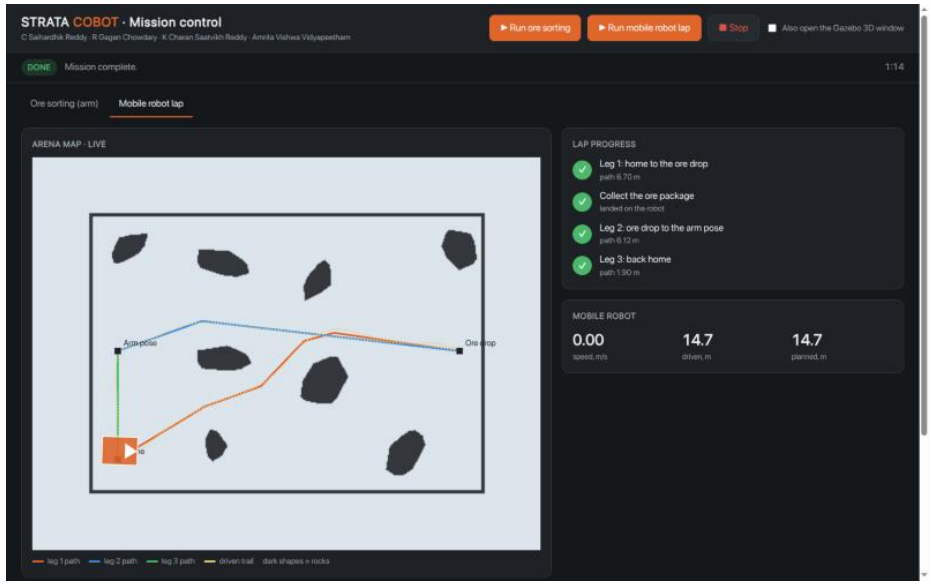
Trial	Outcome
Arm 1–2	4/4 sorted each; one ore 0.83 m away
Arm 3–4	New colour-vote perception: 4/4 each
Robot 1	First planner: 14.74 m vs 14.46 m reference
Robot 2	Final planner: 14.55 m vs 14.58 m ($\eta = 1.00$)
Robot 3	Final planner: 14.48 m vs 14.43 m

No collision and no safety stop in any trial; tracking within 0.09 m.

Live dashboard: one click runs a mission

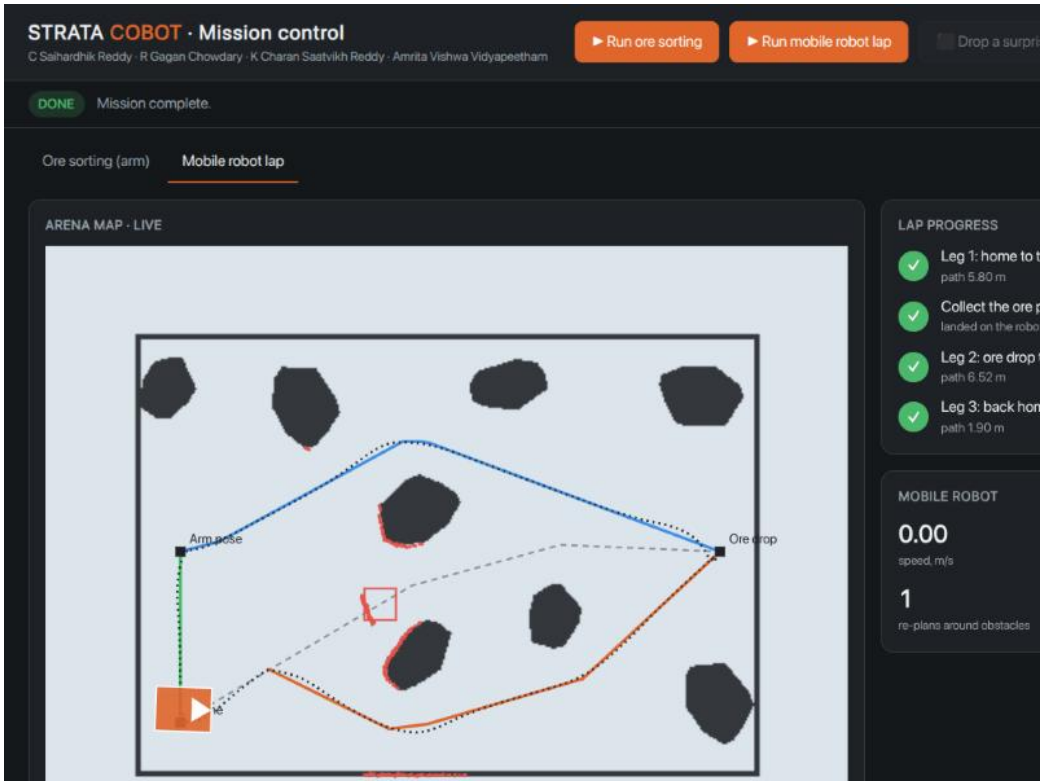


Ore sorting: camera with ore labels, arm step, magnet force, votes



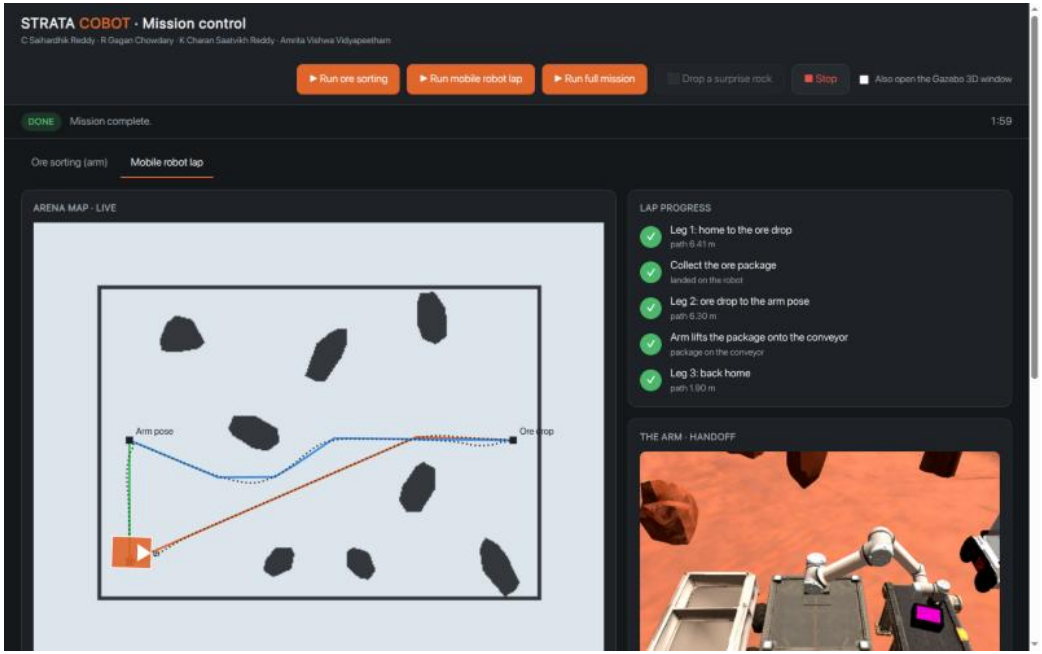
Mobile robot lap: map, planned legs, robot to scale, progress

Re-planning around a rock the map does not have



- Planned paths keep $\geq$0.30 m from every mapped rock
- So a lidar return within 0.26 m of the path ahead is **new**: the eBot stops
- The planner adds it to the map and re-plans from where the eBot stands in 0.1 s
- Demo: a surprise rock dropped mid-leg; lap completed, never closer than 1.16 m

The full mission: haul, hand off, return



1. eBot collects the ore package and drives to the arm
2. Arm finds it (odometry + depth, **4 mm**), picks it: **2.94 N** = 0.30 kg held
3. Lifts it clear of the deck, sets it on the conveyor intake
4. eBot drives home

One click on the dashboard; about 2 minutes end to end.

What broke, and how we fixed it

Problem	**Cause and fix**
No path at all	Drop pose 0.375 m from a wall; full-resolution clearance fixed it
Paths 2% too long	Grid-snapped corners; tightening reached efficiency 1.00
IK took minutes	Checks inside every optimiser step; DLS with one final check
Arm lost after a drop	No IK seed back; a library of solved poses
Vanadinite unreachable	Release at the nearest reachable point
Impure ores	Shape from depth + colour vote, tested on real photos

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Both robots complete their missions end to end

Done

- Sorting: 16/16 ores in the right container
- Planning: paths as short as the reference
- Robust to impure ores: 70% on real photos
- Hand-off: eBot to arm to conveyor
- Re-planning around unmapped rocks

Next

- Train a classifier on real ore samples
- Move to physical robots

Thank you · Questions?